



ZenAI

Local-first Enterprise AI for ZenX

No Claude • No Grok • No paid LLM APIs

Gemma 4 + RAG + Rules + ML + ZenX Data

The New ZenAI Strategy

Remove dependency on paid proprietary models while keeping enterprise-grade intelligence.

LOCAL AI CORE

Gemma 4
Self-hosted inference
No per-query LLM API bill
Private enterprise data



CONTROL LAYER

RAG
Rules Engine
Security & RBAC
Citations & validation



ACTION LAYER

Python/ML
n8n workflows
APIs
Human approval



Principle: the LLM reasons over trusted data; deterministic systems calculate, validate and execute.

ZenAI Architecture

A modular local-first stack that can scale from pilot to enterprise.

USER / APPLICATIONS

ZenX Chat • Dashboards • Reports • Admin



AGENT ORCHESTRATOR

Intent detection • task planning • tool selection • memory



INTELLIGENCE

Gemma 4 • embeddings • RAG • Rules Engine • Guardrails



ANALYTICS

Python • ML • anomaly detection • forecasting • predictive maintenance



AUTOMATION

APIs • alerts • tickets • approvals • Audit Logger



DATA FOUNDATION

PostgreSQL • time-series data • documents • IoT/API/ERP



Deployment Profiles

Three profiles, one architecture — sized to any client's hardware.

LIGHTWEIGHT

CPU-only, 8–16 vCPU, 32–64GB RAM
Gemma 4 E2B/E4B via Ollama
Small sites, air-gapped installs



STANDARD

Single GPU (RTX 4090, L4)
Gemma 4 12B/26B MoE via vLLM
Mid-size sites, moderate concurrency



ENTERPRISE

1–4 GPUs, Kubernetes
Gemma 4 26B MoE/31B via vLLM/NIM
High-volume, multi-site, compliance-heavy



All profiles are fully offline, need no external APIs and run on Apache 2.0 models at zero licensing cost.

VPS Option — Development / POC Only



- Useful for development, proof-of-concept, or small internal testing — not for final client delivery.
- Bind inference to 127.0.0.1 — never 0.0.0.0.
- Firewall: deny all inbound by default; allow only 22, 80, 443. Never expose the inference port directly.
- Reverse proxy with authentication (Basic Auth or Bearer Token) and automatic HTTPS.
- SSH: key-only, password login disabled; brute-force protection enabled.
- Disk encryption on the partition holding model weights.
- Egress firewall rules to prevent outbound data leaks.

Rule: the VPS is a dev/POC environment only — final delivery is client-hosted.

Model Strategy

PRIMARY MODEL

Gemma 4

Use as the main reasoning and generation model.

Self-host through supported local inference stacks.



SPECIALIZED LOCAL MODELS

Use small/open models where they are better suited:

- embeddings
- reranking
- classification
- speech / OCR

Keep model interfaces replaceable.



FUTURE-PROOFING

Create a Model Gateway.

ZenAI can add another self-hosted/open model later without changing product logic.

Paid APIs remain optional, not required.



Why Gemma 4



- Apache 2.0 — free for commercial use, modification and redistribution. No user caps, no royalties.
- Fully offline — no telemetry, no phone-home, no API calls. Data never leaves the client's perimeter.
- Hardware flexible — E2B runs on 3GB RAM; 31B runs on a single high-end GPU.
- Multimodal — text, image, audio, video, with a 256K context window.
- Native tool use — function calling for safe, controlled automation.
- A security-hardened variant (fine-tuned E2B) exists for air-gapped, regulated environments.

The model stays free and auditable — the system gets more capable around it.

The Brain: Agent Orchestrator

Do not let the LLM directly control the whole application.

1. UNDERSTAND

Identify intent
User / tenant
Module
Required data



2. RETRIEVE

RAG
SQL
Time-series
Documents
Policies



3. ANALYZE

Gemma 4
ML models
Rules
Calculations



4. ACT

n8n
API
Alert
Ticket
Approval



5. AUDIT

Evidence
Citations
Logs
Trace



RAG + Rules Engine

The answer must be grounded, validated and auditable.

RAG

Company policies
BRSR / GRI documents
ISO procedures
Equipment manuals
Regulatory documents



RULES ENGINE

Emission-factor checks
Thresholds
Compliance rules
Data-quality checks
Approval rules



OUTPUT

Answer + evidence
Source references
Confidence / data status
Recommended next step



ZenAI Feature Stack

AI capabilities built on the same core architecture.

ZenAI Copilot

Context-aware Q&A with citations



ESG/BRSR Writing Support

Suggest text based on client data & trained over thousands of ESG report



Real-Time AI Insights

ML anomaly detection + explanations



KPI Recommender

Profile + historical data + framework mapping



Regulatory Radar

Source monitoring + impact analysis + action checklist



Regulatory Radar — Local-First Workflow

Automate monitoring without requiring a paid LLM API.

SOURCES

Regulator websites
Official notices
Standards /
circulars
Approved source
list



INGEST

Scheduled collection
Document parsing
Change detection
Source metadata



AI ANALYSIS

Gemma 4 summarizes
RAG grounds claims
Rules classify
impact
Human review
where needed



OUTPUT

Dashboard alert
Email
Action checklist
Audit log



n8n: Automation Layer, Not the AI Brain

ZENAI CORE

Agent orchestration

RAG

Rules

Gemma 4

ML

Security

Audit trail



n8n

Self-hosted workflow automation

API integrations

Notifications

Scheduled jobs

Approvals

Ticketing



WHY THIS DESIGN

Core IP stays inside ZenX.

Automation is replaceable.

No dependency on paid AI APIs.

Clear separation of responsibility.



Security, Governance & Enterprise Control



- Tenant isolation and RBAC
- Private/self-hosted model inference for sensitive deployments
- Prompt injection and document validation controls
- Tool permissions and approval gates
- Every AI answer linked to evidence where applicable
- Full prompt / tool / decision audit logs
- Model gateway so ZenX is not locked to one model
- Evaluation suite using real ZenX customer scenarios

Goal: explainable AI that enterprises can inspect, control and audit.

Security by Construction



- Eliminates entire threat classes by removing internet egress.
- Enforces tenant/site isolation at every layer — data, vector, prompt, retrieval.
- Protects data in transit and at rest with verifiable encryption.
- Verifies model and container integrity via signatures, checksums and SBOMs.
- Guards against AI-specific threats like prompt injection and harmful output.
- Provides tamper-evident, HMAC-chained audit trails for compliance and forensics.
- Applies least-privilege access with strong authentication and RBAC/ABAC.
- Keeps humans in the loop for high-risk actions.
- Maps directly to SOC 2, HIPAA, GDPR, DPDP and EU AI Act Article 12.
- Works fully offline — data never leaves the client's control.

Security isn't bolted on — it's embedded in every layer, in every deployment profile.

Responsibility Split

Who owns what across a client-hosted ZenAI deployment.

Activity	Client	ZenLynx
Infrastructure provisioning (hardware/cloud account)	Responsible	Consulted
Network / firewall configuration	Responsible	Consulted
Application deployment & configuration	Consulted	Responsible
Model updates & tuning	Informed	Responsible
Data ingestion (initial corpus)	Responsible	Consulted
Ongoing operational data feeds	Responsible	Informed
Security / audit review	Consulted	Responsible
Day-to-day monitoring	Consulted	Responsible
Incident response	Consulted	Responsible

ZenAI V2 : Build Sequence

Phase 1

Gemma 4 + FastAPI + PostgreSQL + RAG + citations



Phase 2

Rules Engine + Agent Orchestrator + tool calling



Phase 3

Python ML + ZenEnergy anomaly / predictive maintenance



Phase 4

n8n + alerts + tickets + approvals



Phase 5

Regulatory Radar + ESG/BRSR automation



Phase 6

Model Gateway + evaluation + enterprise hardening



Target architecture: 100% usable without Claude/Grok or any paid LLM API. Infrastructure and compute still have normal hosting/GPU costs.